ACYCLICITY OF THE PUSHFORWARD OF AN INJECTIVE SHEAF FOR THE LERAY SPECTRAL SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all three clauses of the statement.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Pushforward of an injective abelian sheaf along a morphism of schemes and its Cech cohomology. Same chapter, the pushforward f_*I of an injective abelian sheaf and the computation of Cech cohomology of injective sheaves. Supplies the object whose acyclicity is asserted.
- Acyclicity criteria for the sections functor and for the pushforward. Same chapter, the criterion by vanishing of Cech cohomology over a cofinal system of coverings, applied to the functor Gamma(V, -) and to g_*. Supplies the two acyclicity conclusions.

Source reference. Original lemma 03OR: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1702–1702. Navigation pointer invoked at proof line 8160; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03F9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8165–8165. Navigation pointer invoked at proof line 8165; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 03Q8: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7510–7510. Navigation pointer invoked at proof line 8168; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : X \to Y$ be a morphism and $\mathcal{I}$ an injective object of $Ab(X_{étale})$. Let $V \in \mathrm{Ob}(Y_{étale})$. Then*

(1) *for any covering $\mathcal{V} = \{V_j \to V\}_{j \in J}$ we have $\check{H}^p(\mathcal{V}, f_*\mathcal{I}) = 0$ for all $p > 0$,*
(2) *$f_*\mathcal{I}$ is acyclic for the functor $\Gamma(V, -)$, and*
(3) *if $g : Y \to Z$, then $f_*\mathcal{I}$ is acyclic for g_*.*

Proof. Observe that $\check{\mathcal{C}}^\bullet(\mathcal{V}, f_*\mathcal{I}) = \check{\mathcal{C}}^\bullet(\mathcal{V} \times_Y X, \mathcal{I})$ which has vanishing higher cohomology groups by Lemma 03OR. This proves (1). The second statement follows as a sheaf which has vanishing higher Čech cohomology groups for any covering has vanishing higher cohomology groups. This a wonderful exercise in using the Čech-to-cohomology spectral sequence, but see Cohomology on Sites, Lemma 03F9 for details and a more precise and general statement. Part (3) is a consequence of (2) and the description of $R^p g_*$ in Lemma 03Q8. □